

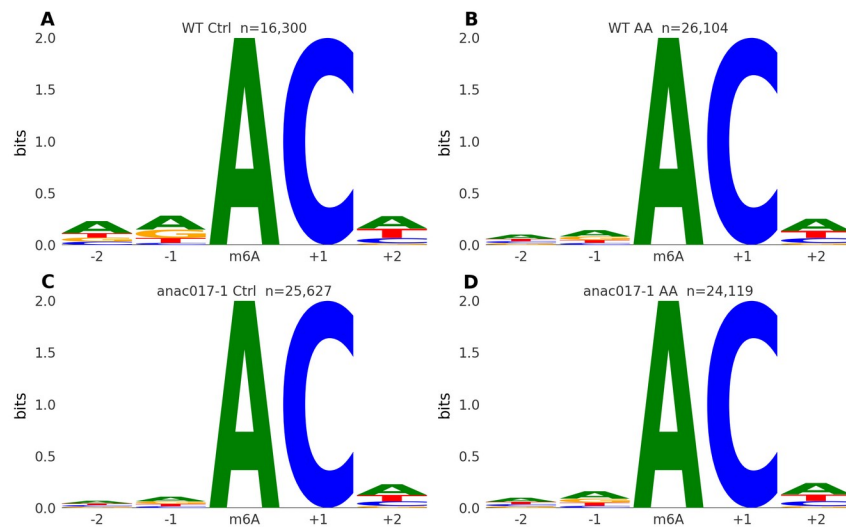
K-CHOPORE

A reproducible nanopore Direct RNA-seq
pipeline for the plant m6A
epitranscriptome

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The gap

- **m6A** is the most abundant mRNA modification — it tunes stability, splicing, translation.
- Nanopore **Direct RNA-seq (DRS)** reads it on **native** RNA.
- But plant-ready, **reproducible** workflows are scarce.



K-CHOPORE — signal →

The pipeline



One Snakemake config controls the whole run;
precomputed steps are reused, only what changed
re-runs.

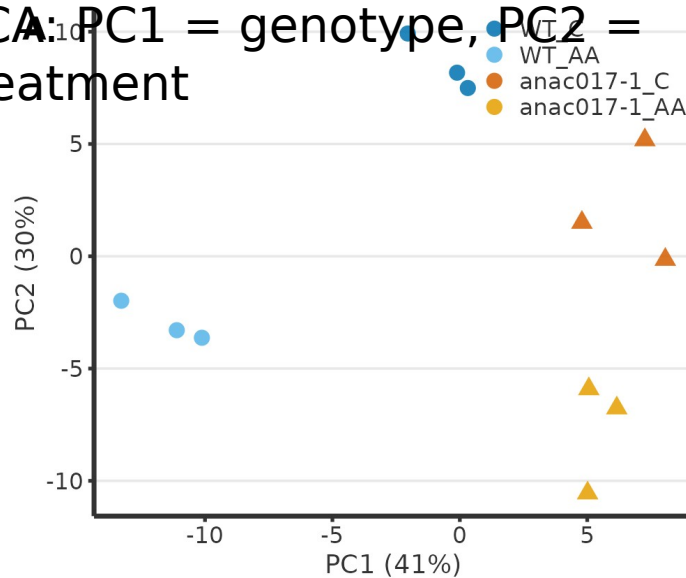
Case study: mitochondrial retrograde response

2x2 design · *WT* vs *anac017* · response

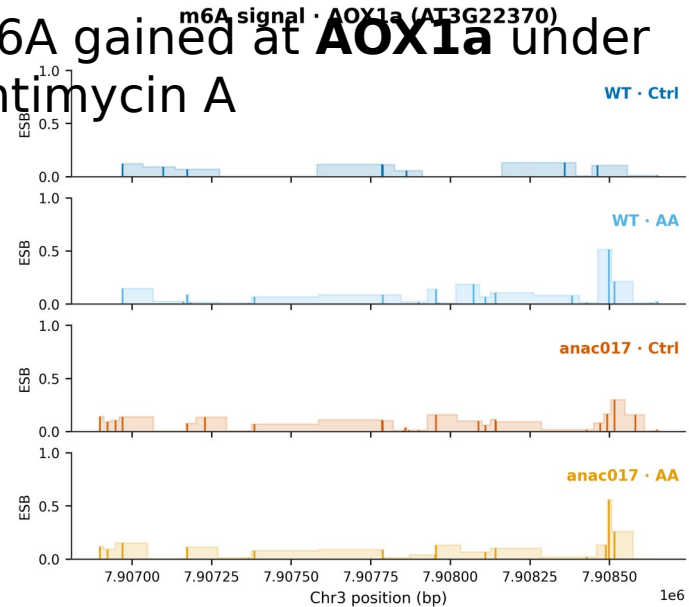
1 × Control vs Antimycin A · n

= 3

PCA: PC1 = genotype, PC2 = treatment

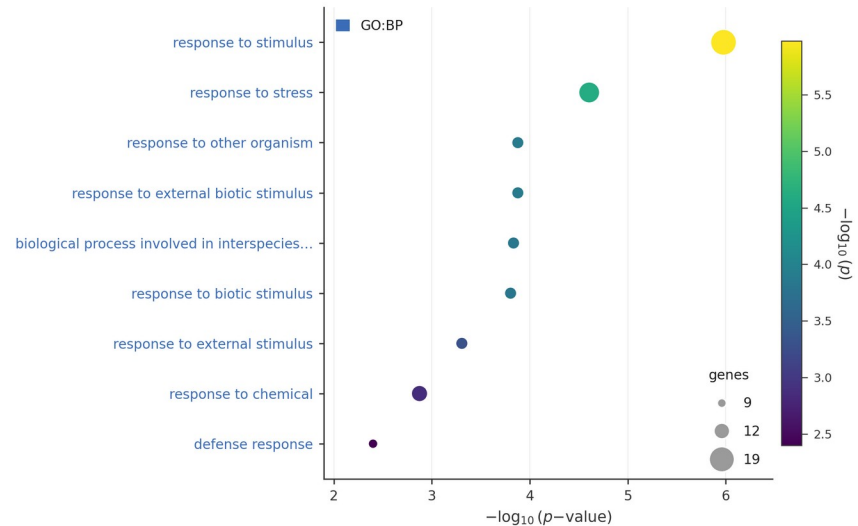


m6A gained at *AOX1a* under Antimycin A



What we find

- **175** transcripts respond to AA; **32** in an **ANAC017-dependent** way (interaction).
- ~**89%** of AA-gained m6A sites in WT are **ANAC017-dependent**.
- AA-responsive genes enriched in **stress / hypoxia** (e.g. *CYP81D8*).



Reproducible by design

Docker image + Snakemake + one config

Clone → `docker run` → edit config → run.

github.com/biopelayo/kchopore-anac017-drs

*A reusable, FAIR resource for plant
epitranscriptomics.*

Thank you

K-CHOPORE turns raw nanopore signal into a coverage-controlled m6A map — reproducibly.

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